



Indolent B-Cell Non-Hodgkin Lymphoma Treatment (PDQ®)–Health Professional Version

General Information About B-Cell Non-Hodgkin Lymphoma

The non-Hodgkin lymphomas (NHL) are a heterogeneous group of lymphoproliferative malignancies with differing patterns of behavior and responses to treatment.[1] This summary focuses primarily on indolent B-cell NHL. For information about B-cell and T-cell lymphomas, see [Aggressive B-Cell Non-Hodgkin Lymphoma Treatment](#), [Peripheral T-Cell Non-Hodgkin Lymphoma Treatment](#) and [Mycosis Fungoides and Other Cutaneous T-Cell Lymphomas Treatment](#).

Like Hodgkin lymphoma, NHL usually originates in lymphoid tissues and can spread to other organs. However, NHL is much less predictable than Hodgkin lymphoma and has a far greater tendency to spread to extranodal sites. The prognosis depends on the histological type, disease stage, and treatment.

Incidence and Mortality

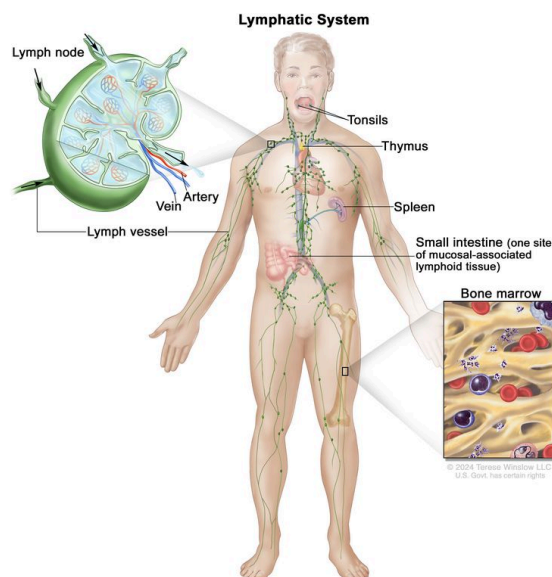
Estimated new cases and deaths from all types of NHL in the United States in 2025:[2]

- New cases: 80,350.
- Deaths: 19,390.

B-cell lymphomas make up about 85% of NHL cases.[3]

Anatomy

NHL usually originates in lymphoid tissues.



The lymph system is part of the body's immune system and is made up of tissues and organs that help protect the body from infection and disease. These include the tonsils, adenoids (not shown), thymus, spleen, bone marrow, lymph vessels, and lymph nodes. Lymph tissue is also found in many other parts of the body, including the small intestine.

Prognosis and Survival

NHL can be divided into two prognostic groups: indolent lymphomas and aggressive lymphomas.

Indolent NHL has a relatively good prognosis, with a median survival as long as 20 years, but it is usually not curable in advanced clinical stages.[4] Early-stage (stage I and stage II) indolent NHL can be effectively treated with radiation therapy alone. Most of the indolent NHLs are nodular (or follicular) in morphology.

Aggressive NHL has a worse prognosis in the short term, but a significant number of patients can be cured with intensive combination chemotherapy regimens.

More than 70% of patients with aggressive NHL can be cured. Most relapses occur in the first 2 years after therapy. The risk of late relapse is higher in patients who manifest both indolent and aggressive histologies.[5]

Although indolent NHL responds quickly to immunotherapy, radiation therapy, and chemotherapy, a continuous rate of relapse is usually seen in advanced stages. However, patients can often be re-treated with considerable success if the disease histology remains low grade. Patients who present with, or convert to, aggressive forms of NHL may have sustained complete remissions with combination chemotherapy regimens or aggressive consolidation with marrow or stem cell support.[6,7]

Late Effects of Treatment of NHL

Late effects of treatment of non-Hodgkin lymphoma (NHL) have been observed. Impaired fertility may occur after exposure to alkylating agents.[8] For as many as three decades after diagnosis, patients are at a significantly elevated risk of developing second primary cancers, especially the following:[9-12]

- Lung cancer.
- Brain cancer.
- Kidney cancer.
- Bladder cancer.
- Melanoma.
- Hodgkin lymphoma.
- Acute nonlymphocytic leukemia.

Left ventricular dysfunction was a significant late effect in long-term survivors of high-grade NHL who received more than 200 mg/m² of doxorubicin.[8,13]

Myelodysplastic syndrome and acute myelogenous leukemia are late complications of myeloablative therapy with autologous bone marrow or peripheral blood stem cell support, as well as conventional chemotherapy-containing alkylating agents.[10,14-21] Most of these patients show clonal hematopoiesis even before the transplant, suggesting that the hematologic injury usually occurs during induction or reinduction chemotherapy.[16,22,23] A series of 605 patients who received autologous bone marrow transplant (BMT) with cyclophosphamide and total-body radiation therapy (as conditioning) were followed for a median of 10 years. The incidence of a second malignancy was 21%, and 10% of those malignancies were solid tumors.[24]

A study of young women who received autologous BMT reported successful pregnancies with children born free of congenital abnormalities.[25] Late-occurring venous thromboembolism can occur after allogeneic or autologous BMT.[26]

Some patients have osteopenia or osteoporosis at the start of therapy; bone density may worsen after therapy for lymphoma.[27]

Long-term impaired immune health was evaluated in a retrospective cohort study of 21,690 survivors of diffuse large B-cell lymphoma from the California Cancer Registry. Elevated incidence rate ratios were found up to 10 years later for pneumonia (10.8-fold), meningitis (5.3-fold), immunoglobulin deficiency (17.6-fold), and autoimmune cytopenias (12-fold).[28] Similarly, there are impaired humoral responses to COVID-19 virus vaccination in patients with lymphoma who receive B-cell-directed therapies.[29,30]

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Cellular Classification of B-Cell Non-Hodgkin Lymphoma

A pathologist should be consulted before a biopsy because some studies require special preparation of tissue (e.g., frozen tissue). Knowledge of cell surface markers and immunoglobulin and T-cell receptor gene rearrangements may help with diagnostic and therapeutic decisions. The clonal excess of light-chain immunoglobulin may differentiate malignant cells from reactive cells. Because the prognosis and the approach to treatment are influenced by histopathology, outside biopsy specimens should be carefully reviewed by a

hematopathologist who is experienced in diagnosing lymphomas. Although lymph node biopsies are recommended whenever possible, sometimes immunophenotypic data are sufficient for diagnosis of lymphoma when fine-needle aspiration cytology or core needle biopsy is preferred.^[1,2]

Historical Classification Systems

Historically, uniform treatment of patients with non-Hodgkin lymphoma (NHL) has been hampered by the lack of a uniform classification system. In 1982, results of a consensus study were published as the Working Formulation.^[3] The Working Formulation combined results from six major classification systems into one classification. This allowed comparison of studies from different institutions and countries. The Rappaport classification, which also follows, is no longer in common use.

Table 1. Historical Classification Systems for Non-Hodgkin Lymphoma (NHL)

Working Formulation ^[3]	Rappaport Classification
Low grade	
A. Small lymphocytic, consistent with chronic lymphocytic leukemia	Diffuse lymphocytic, well-differentiated
B. Follicular, predominantly small-cleaved cell	Nodular lymphocytic, poorly differentiated
C. Follicular, mixed small-cleaved, and large cell	Nodular mixed, lymphocytic, and histiocytic
Intermediate grade	
D. Follicular, predominantly large cell	Nodular histiocytic
E. Diffuse small-cleaved cell	Diffuse lymphocytic, poorly differentiated
F. Diffuse mixed, small and large cell	Diffuse mixed, lymphocytic, and histiocytic

Working Formulation [3]	Rappaport Classification
G. Diffuse, large cell, cleaved, or noncleaved cell	Diffuse histiocytic
High grade	
H. Immunoblastic, large cell	Diffuse histiocytic
I. Lymphoblastic, convoluted, or nonconvoluted cell	Diffuse lymphoblastic
J. Small noncleaved-cell, Burkitt, or non-Burkitt	Diffuse undifferentiated Burkitt or non-Burkitt

Current Classification Systems

As the histopathological diagnosis of NHL has become more sophisticated with the use of immunologic and genetic techniques, a number of new pathological entities have been described.[4] In addition, the understanding and treatment of many of the previously described pathological subtypes have changed. As a result, the Working Formulation has become outdated and less useful to clinicians and pathologists. European and American pathologists have proposed a new classification, the Revised European American Lymphoma (REAL) classification.[5-8] Since 1995, members of the European and American Hematopathology societies have been collaborating on a new World Health Organization (WHO) classification, which represents an updated version of the REAL system.[9,10]

Updated REAL/WHO classification

The World Health Organization (WHO) modification of the Revised European American Lymphoma (REAL) classification recognizes three major categories of lymphoid malignancies based on morphology and cell lineage: B-cell neoplasms, T-cell/natural killer (NK)-cell neoplasms, and Hodgkin lymphoma (HL). Both lymphomas and lymphoid leukemias are included in this classification because both solid and circulating phases are present in many lymphoid neoplasms and distinction between them is artificial. For example, B-cell chronic lymphocytic leukemia (CLL) and B-cell small lymphocytic lymphoma are simply different manifestations of the same neoplasm, as are lymphoblastic lymphomas and acute lymphocytic leukemias. Within the B-cell and T-cell categories, two subdivisions are recognized: precursor neoplasms, which correspond to the earliest stages of differentiation, and more mature differentiated neoplasms.[9,10]

B-cell neoplasms

1. Precursor B-cell neoplasm: precursor B-acute lymphoblastic leukemia/lymphoblastic lymphoma (LBL).
2. Peripheral B-cell neoplasms.
 - a. B-cell CLL/small lymphocytic lymphoma.
 - b. B-cell prolymphocytic leukemia.
 - c. Lymphoplasmacytic lymphoma/immunocytoma.
 - d. Mantle cell lymphoma.
 - e. Follicular lymphoma.
 - f. Extranodal marginal zone B-cell lymphoma of mucosa-associated lymphatic tissue (MALT) type.
 - g. Nodal marginal zone B-cell lymphoma ($\pm$ monocytoid B cells).
 - h. Splenic marginal zone lymphoma ($\pm$ villous lymphocytes).
 - i. Hairy cell leukemia.
 - j. Plasmacytoma/plasma cell myeloma.
 - k. Diffuse large B-cell lymphoma.
 - l. Burkitt lymphoma.

T-cell and putative NK-cell neoplasms

1. Precursor T-cell neoplasm: precursor T-acute lymphoblastic leukemia/LBL. For more information, see [Acute Lymphoblastic Leukemia Treatment](#).
2. Peripheral T-cell and NK-cell neoplasms.
 - a. T-cell CLL/prolymphocytic leukemia.
 - b. T-cell granular lymphocytic leukemia.
 - c. Mycosis fungoides (including Sézary syndrome).
 - d. Peripheral T-cell lymphoma, not otherwise specified.
 - e. Hepatosplenic gamma/delta T-cell lymphoma.
 - f. Subcutaneous panniculitis-like T-cell lymphoma.
 - g. Extranodal T-/NK-cell lymphoma, nasal type.
 - h. Nodal lymphomas of T follicular helper cell origin (including angioimmunoblastic T-cell lymphoma, follicular peripheral T-cell lymphoma, and nodal peripheral T-cell lymphoma with T follicular helper phenotype).

- i. Enteropathy-associated intestinal T-cell lymphoma.
- j. Monomorphic epitheliotropic intestinal T-cell lymphoma.
- k. Adult T-cell lymphoma/leukemia (human T-lymphotrophic virus [HTLV] 1+).
- l. Anaplastic large cell lymphoma, primary systemic type.
- m. Anaplastic large cell lymphoma, primary cutaneous type.
- n. Aggressive NK-cell leukemia.

HL

- 1. Nodular lymphocyte-predominant HL.
- 2. Classical HL.
 - a. Nodular sclerosis HL.
 - b. Lymphocyte-rich classical HL.
 - c. Mixed-cellularity HL.
 - d. Lymphocyte-depleted HL.

The REAL classification encompasses all lymphoproliferative neoplasms. For more information, see the following PDQ summaries:

- [Acute Lymphoblastic Leukemia Treatment](#)
- [Aggressive B-Cell Non-Hodgkin Lymphoma Treatment](#)
- [AIDS-Related Lymphoma Treatment](#)
- [Chronic Lymphocytic Leukemia Treatment](#)
- [Hodgkin Lymphoma Treatment](#)
- [Hairy Cell Leukemia Treatment](#)
- [Indolent B-Cell Non-Hodgkin Lymphoma Treatment](#)
- [Mantle Cell Lymphoma Treatment](#)
- [Mycosis Fungoides and Other Cutaneous T-Cell Lymphomas Treatment](#)
- [Peripheral T-Cell Non-Hodgkin Lymphoma Treatment](#)
- [Plasma Cell Neoplasms \(Including Multiple Myeloma\) Treatment](#)
- [Primary Central Nervous System Lymphoma Treatment](#)

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Stage Information for Indolent B-Cell Non-Hodgkin Lymphoma

Stage is important in selecting a treatment for patients with non-Hodgkin lymphoma (NHL). Chest and abdominal computed tomography (CT) scans are usually part of the staging evaluation for all patients with lymphoma. The staging system for NHL is similar to the staging system used for Hodgkin lymphoma (HL).

It is common for patients with NHL to have involvement of the following sites:

- Noncontiguous lymph nodes.
- Waldeyer ring.
- Epitrochlear nodes.
- Gastrointestinal tract.
- Extranodal presentations. (A single extranodal site is occasionally the only site of involvement in patients with diffuse lymphoma.)
- Bone marrow.
- Liver (especially common in patients with low-grade lymphomas).

Cytological examination of cerebrospinal fluid may be positive in patients with aggressive NHL. Involvement of hilar and mediastinal lymph nodes is less common than in HL. Mediastinal adenopathy, however, is a prominent feature of lymphoblastic lymphoma and primary mediastinal B-cell lymphoma, entities primarily found in young adults.

Most patients with NHL present with advanced (stage III or stage IV) disease often identified by CT scans or biopsies of the bone marrow and other accessible sites of involvement. In a retrospective review of over 32,000 cases of lymphoma in France, up to 40% of diagnoses were made by core needle biopsy, and 60% were made by excisional biopsy.[1] After expert review, core needle biopsy provided a definite diagnosis in 92.3% of cases; excisional biopsy provided a definite diagnosis in 98.1% of cases ($P < .0001$). Laparoscopic biopsy or laparotomy is not required for staging but rarely may be necessary to establish a diagnosis or histological type.[2]

Positron emission tomography (PET) with fluorine F 18-fludeoxyglucose can be used for initial staging. It can also be used for follow-up after therapy as a supplement to CT scanning.[3] Multiple studies have demonstrated that interim PET scans after two to four cycles of therapy do not provide reliable prognostic information. A large cooperative group trial ([ECOG-E344 \[NCT00274924\]](#)) reported problems with interobserver reproducibility. Two prospective trials and one meta-analysis showed no differences in outcomes between PET-negative and PET-positive/biopsy-negative patients.[4-7]

In a retrospective study of 130 patients with diffuse large B-cell lymphoma, PET scanning identified all clinically important marrow involvement from lymphoma, and bone marrow

biopsy did not upstage any patient's lymphoma.[8] A retrospective study of 580 patients with follicular lymphoma from seven National Cancer Institute–sponsored trials showed no improvement in assessing response to therapy when bone marrow biopsy was added to radiological imaging.[9] The workup of NHL should include bone marrow biopsy when management would change (e.g., determining limited stage vs. advanced stage) or when evaluating cytopenias.

For patients with follicular lymphoma, a positive PET result after therapy has a worse prognosis; however, it is unclear whether a positive PET result is predictive when further or different therapy is implemented.[10]

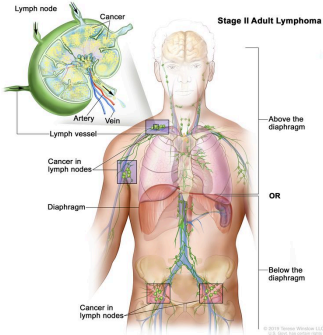
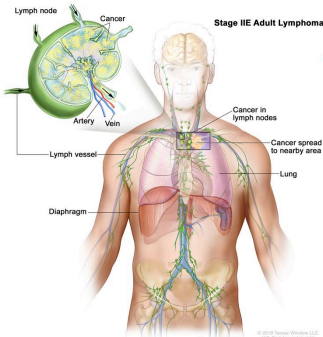
Staging Subclassification System

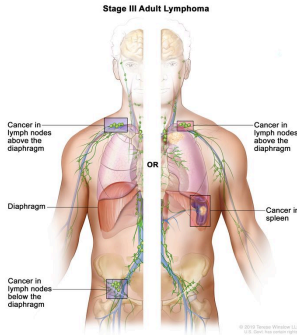
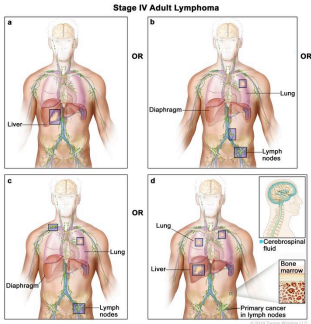
Lugano classification

The American Joint Committee on Cancer (AJCC) has adopted the Lugano classification to evaluate and stage lymphoma.[11] The Lugano classification system replaces the Ann Arbor classification system, which was adopted in 1971 at the Ann Arbor Conference,[12] with some modifications 18 years later from the Cotswolds meeting.[13,14]

Table 2. Lugano Classification for Hodgkin and Non-Hodgkin Lymphoma^a

Stage	Stage Description	Illustration
<i>Limited stage</i>		
I	Involvement of a single lymphatic site (i.e., nodal region, Waldeyer’s ring, thymus, or spleen).	

Stage	Stage Description	Illustration
IE	Single extralymphatic site in the absence of nodal involvement (rare in Hodgkin lymphoma).	
II	Involvement of two or more lymph node regions on the same side of the diaphragm.	 The diagram illustrates Stage II Adult Lymphoma. It features a human torso with lymph nodes highlighted. Labels indicate 'Cancer in lymph nodes' in two separate regions, one above the diaphragm and one below. A vertical line marks the diaphragm. To the right, text reads 'Stage II Adult Lymphoma' and 'Above the diaphragm' and 'Below the diaphragm' with an 'OR' between them. An inset shows a lymph node with cancer cells, an artery, a vein, and a lymph vessel.
IIE	Contiguous extralymphatic extension from a nodal site with or without involvement of other lymph node regions on the same side of the diaphragm.	 The diagram illustrates Stage IIE Adult Lymphoma. It features a human torso with lymph nodes highlighted. Labels indicate 'Cancer in lymph nodes' in one region and 'Cancer spread to nearby area' pointing to the lung. A vertical line marks the diaphragm. To the right, text reads 'Stage IIE Adult Lymphoma'. An inset shows a lymph node with cancer cells, an artery, a vein, and a lymph vessel.
II _{bulky} ^b	Stage II with disease bulk. ^c	
Advanced stage		

Stage	Stage Description	Illustration
III	Involvement of lymph node regions on both sides of the diaphragm; nodes above the diaphragm with spleen involvement.	 The diagram, titled "Stage III Adult Lymphoma", shows a human torso with lymphatic nodes highlighted. Labels indicate "Cancer in lymph nodes above the diaphragm" (upper chest/neck area) and "Cancer in lymph nodes below the diaphragm" (lower chest/abdomen area). A label "Diaphragm" points to the horizontal line separating the upper and lower regions. Another label "Cancer in spleen" points to the spleen in the abdominal area. The word "OR" is placed between the two main lymph node regions, indicating that either side can be involved.
IV	Diffuse or disseminated involvement of one or more extralymphatic organs, with or without associated lymph node involvement; or noncontiguous extralymphatic organ involvement in conjunction with nodal stage II disease; or any extralymphatic organ involvement in nodal stage III disease. Stage IV includes any involvement of the CSF, bone marrow, liver, or multiple lung lesions (other than by direct extension in stage IIE disease).	 The diagram, titled "Stage IV Adult Lymphoma", shows four scenarios (a, b, c, d) separated by "OR" labels, illustrating different patterns of extralymphatic involvement. a) Shows involvement of the liver and lymph nodes. b) Shows involvement of the lung, diaphragm, and lymph nodes. c) Shows involvement of the lung, diaphragm, and lymph nodes. d) Shows involvement of the lung, liver, and lymph nodes, with a callout for "Primary cancer in lymph nodes". Additional callouts in the diagrams include "Cerebrospinal fluid" and "Bone marrow".

Note: Hodgkin lymphoma uses A or B designation with stage group. A/B is no longer used in NHL.

Stage	Stage Description	Illustration
CSF = cerebrospinal fluid; CT = computed tomography; DLBCL = diffuse large B-cell lymphoma; NHL = non-Hodgkin lymphoma. ^a Hodgkin and Non-Hodgkin Lymphomas. In: Amin MB, Edge SB, Greene FL, et al., eds.: <i>AJCC Cancer Staging Manual</i> . 8th ed. New York, NY: Springer, 2017, pp. 937–58. ^b Stage II bulky may be considered either early or advanced stage based on lymphoma histology and prognostic factors. ^c The definition of disease bulk varies according to lymphoma histology. In the Lugano classification, bulk in Hodgkin lymphoma is defined as a mass greater than one-third of the thoracic diameter on CT of the chest or a mass >10 cm. For NHL, the recommended definitions of bulk vary by lymphoma histology. In follicular lymphoma, 6 cm has been suggested based on the Follicular Lymphoma International Prognostic Index-2 and its validation. In DLBCL, cutoffs ranging from 5 cm to 10 cm have been used, although 10 cm is recommended.		

Occasionally, specialized staging systems are used. The physician should be aware of the system used in a specific report.

The E designation is used when extranodal lymphoid malignancies arise in tissues separate from, but near, the major lymphatic aggregates. Stage IV refers to disease that is diffusely spread throughout an extranodal site, such as the liver. If pathological proof of involvement of one or more extralymphatic sites has been documented, the symbol for the site of involvement, followed by a plus sign (+), is listed.

Table 3. Notation to Identify Specific Sites

N = nodes	H = liver	L = lung	M = bone marrow
S = spleen	P = pleura	O = bone	D = skin

Current practice assigns a clinical stage based on the findings of the clinical evaluation and a pathological stage based on the findings from invasive procedures beyond the initial biopsy.

For example, on percutaneous biopsy, a patient with inguinal adenopathy and a positive lymphangiogram without systemic symptoms might have involvement of the liver and bone marrow. The precise stage of such a patient would be clinical stage IIA, pathological stage IVA(H+)(M+).

Several other factors that are not included in the above staging system are important for the staging and prognosis of patients with NHL. These factors include the following:

- Age.
- Performance status (PS).
- Tumor size.
- Lactate dehydrogenase (LDH) values.
- The number of extranodal sites.

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Indolent B-Cell Non-Hodgkin Lymphoma

Indolent B-cell non-Hodgkin lymphoma (NHL) includes the following subtypes:

- Follicular lymphoma (grades 1–3a).
- Lymphoplasmacytic lymphoma (Waldenström macroglobulinemia).
- Marginal zone lymphoma.

Follicular Lymphoma (Grades 1–3a)

Follicular lymphoma makes up 20% of all NHL and as many as 70% of the indolent lymphomas reported in American and European clinical trials.[\[1-3\]](#) Most patients with follicular lymphoma are aged 50 years and older and present with widespread disease at diagnosis. Nodal involvement is most common and is often accompanied by splenic and bone marrow disease. Rearrangement of the *BCL2* gene is present in more than 90% of patients with follicular lymphoma. Overexpression of the BCL2 protein is associated with the inability to eradicate the lymphoma by inhibiting apoptosis.[\[4\]](#)

Prognosis

Follicular lymphoma is designated as indolent because median survival ranges from 8 to 15 years, even in advanced stages.[\[5-7\]](#) Patients with advanced-stage follicular lymphoma are not cured with current therapeutic options. The rate of relapse is fairly consistent over time, even

in patients who have achieved complete responses to treatment.[8] Watchful waiting, the deferring of treatment until the patient becomes symptomatic, is an option for patients with advanced-stage follicular lymphoma.[9,10] An international index for follicular lymphoma (the Follicular Lymphoma International Prognostic Index [FLIPI]) [11-13] identified five significant risk factors prognostic of overall survival (OS):

1. Age (≤ 60 years vs. > 60 years).
2. Serum lactate dehydrogenase (LDH) (normal vs. elevated).
3. Stage (stage I or stage II vs. stage III or stage IV).
4. Hemoglobin level (≥ 120 g/L vs. < 120 g/L).
5. Number of nodal areas (≤ 4 vs. > 4).

Patients with zero or one risk factor have a 10-year survival rate of 67%, and four or five risk factors confer a 10-year survival rate of 36%.[11] In a revised FLIPI-2, an elevated beta-2-microglobulin and lymph node size of more than 6 cm are proposed prognostic factors instead of serum LDH and the number of nodal areas.[14] Although the FLIPI and FLIPI-2 indices can predict progression-free survival (PFS) and OS, the scores cannot be used to establish the need for therapy or to predict response to therapy.[11,14] The primary use of FLIPI or FLIPI-2 is to assure a balance of prognostic factors or to define entry requirements in randomized clinical trials. Individuals with an adverse FLIPI score may well benefit from watchful waiting or may respond well to initial therapy. An alternative prognostic index using only beta-2-microglobulin and initial bone marrow involvement (PRIMA-PI) has the disadvantage of requiring an invasive test not usually required outside the context of a clinical trial.[15] The Follicular Lymphoma Evaluation Index (FLEX), an alternative prognostic index using only noninvasive clinical variables, outperformed FLIPI, FLIPI-2, and PRIMA-PI, using data from immunochemotherapy trials.[16] The nine clinical variables of the FLEX model include:

1. Male sex.
2. Histological grade 3a disease.
3. More than two extranodal sites.
4. Eastern Cooperative Oncology Group performance status score of 2 or higher.
5. Hemoglobin level less than 120 g/L.
6. Beta-2 microglobulin level above the upper limit of normal.
7. LDH level above the upper limit of normal.
8. Absolute blood natural killer cell count less than 100/ μ L.
9. Bulky disease.

Three retrospective analyses, including one pooled analysis of 5,225 patients from 13 randomized clinical trials, identified a high-risk group that had a 5-year OS rate of 50% when relapses occurred within 24 months of induction chemoimmunotherapy.[17-19] A fourth retrospective analysis of 296 patients who received bendamustine and rituximab found a 2-year OS rate of 38% (95% confidence interval [CI], 20%–55%) among those with progression of disease before 24 months (POD24). Most of these patients (76%) had transformed disease (histological progression to diffuse large B-cell lymphoma [DLBCL]).[20] These patients with higher-risk POD24 disease represent a target population for clinical trials.

Therapeutic approaches

Because of the often-indolent clinical course and the lack of symptoms in some patients with follicular lymphoma, watchful waiting remains a standard of care during the initial encounter and for patients with slow asymptomatic relapsing disease. When therapy is required, numerous options may be used in varying sequences with an OS equivalence at 5 to 10 years.[9,21-23] Rituximab can be given alone or in combination with various chemotherapy options.[23-25] Rituximab can also be combined with the immunomodulating-agent lenalidomide to avoid the short- and long-term toxicities of cytotoxic agents.[26-28] Obinutuzumab is a completely humanized anti-CD20 monoclonal antibody that can be given alone or with combination chemotherapy. It may be especially useful for patients who develop severe allergic reactions to rituximab due to human antimouse antibodies.[29] CD19-directed chimeric antigen receptor T cells may be used in patients who have disease progression after two or more prior lines of therapy.[30] Mosunetuzumab, a bispecific CD20-directed CD3 T-cell engager, may also be used in this setting.[31] Consolidation therapy for relapsed disease after reinduction therapy using autologous stem cell transplant (SCT) or allogeneic SCT can be considered.[32]

Outside the context of clinical trials, the use of measurable residual disease (MRD) testing has not been shown to be predictive in directing therapy for patients with follicular lymphoma. In retrospective analyses of two randomized prospective trials, while MRD negativity was prognostic of outcome, maintenance rituximab or obinutuzumab prolonged PFS the most among patients with MRD-negative disease.[33,34][Level of evidence C2] Stopping maintenance rituximab or obinutuzumab was not indicated in patients with MRD-negative disease, negating any possible change in therapy based on that status.

Follicular lymphoma *in situ* and primary follicular lymphoma of the duodenum are particularly indolent variants that rarely progress or require therapy.[35,36] A so-called pediatric-type nodal follicular lymphoma has indolent behavior and rarely recurs; adult patients with this histological variant have disease characterized by a lack of *BCL2* rearrangement in conjunction with a Ki-67 proliferation index greater than 30% and a localized stage I presentation.[37]

Patients with indolent lymphoma may experience a relapse with a more aggressive histology. If the clinical pattern of relapse suggests that the disease is behaving in a more aggressive manner, a biopsy can be performed, if feasible.[38] If disease conversion to a more aggressive histology is confirmed, therapy must change to a regimen applicable to that histological type.[39] Rapid growth or discordant growth between various disease sites may indicate a histological conversion.[38]

In a prospective nonrandomized study, at a median follow-up of 6.8 years, 379 of 2,652 patients (14%) subsequently transformed to a more aggressive histology after an initial diagnosis of follicular lymphoma.[40][[Level of evidence C3](#)] The median OS after subsequent transformation was 5 years. However, among 47 patients with evidence of transformation in conjunction with follicular lymphoma at the time of initial diagnosis, the OS was no worse than that of the nontransformed patients (5-year OS rate, 88%; 95% CI, 74%–95%).

Grade 3b follicular lymphoma is managed similarly to DLBCL. For more information, see [Aggressive B-Cell Non-Hodgkin Lymphoma Treatment](#).

Lymphoplasmacytic Lymphoma (Waldenström Macroglobulinemia)

Lymphoplasmacytic lymphoma is usually associated with a monoclonal serum paraprotein of immunoglobulin M (IgM) type (Waldenström macroglobulinemia).[41] Most patients have bone marrow, lymph node, and splenic involvement, and some patients may develop hyperviscosity syndrome. Most patients with Waldenström macroglobulinemia carry the *MYD88* variant, which some pathologists consider indicative for the disease.[42] Other lymphomas may also be associated with serum paraproteins. Patients with lymphoplasmacytic lymphoma should be checked for associated hepatitis C virus (HCV) infection.

Asymptomatic patients can be monitored for evidence of disease progression without immediate need for chemotherapy.[9,43,44]

Prognostic factors associated with symptoms requiring therapy include:

- Aged 70 years or older.
- Beta-2-microglobulin of 3 mg/dL or more.
- Increased serum LDH.[43]

An externally validated prognostic model uses age, albumin, and LDH levels.[45]

Therapeutic approaches

The management of lymphoplasmacytic lymphoma is similar to that of other low-grade lymphomas, especially diffuse small lymphocytic lymphoma/chronic lymphocytic leukemia.[46-

48] If the serum viscosity relative to water is greater than four, the patient may have symptoms of hyperviscosity. Plasmapheresis is useful for temporary, acute symptoms such as retinopathy, congestive heart failure, and central nervous system (CNS) dysfunction. It can also be combined with chemotherapy for prolonged disease control. Symptomatic patients with a serum viscosity of four or lower are usually treated with chemoimmunotherapy or biologically directed therapies. Therapy may be required to correct hemolytic anemia in patients with chronic cold agglutinin disease; rituximab, bendamustine, and steroids are often used.[44] Occasionally, a heated room is required for patients whose cold agglutinins become activated by even minor chilling. Sutimlimab, an immunoglobulin G4 monoclonal antibody that selectively inhibits the complement pathway at C15, can reduce hemolysis when therapies directed at the lymphoplasmacytic lymphoma are ineffective.[49]

First-line regimens include zanubrutinib (a Bruton tyrosine kinase [BTK] inhibitor), rituximab, and ibrutinib (another BTK inhibitor), rituximab alone, the nucleoside analogues, and alkylating agents, either as single agents or as part of combination chemotherapy.[50-53] In a randomized prospective trial, 150 symptomatic patients (including patients with previously untreated and relapsing disease) received either ibrutinib and rituximab or rituximab and a placebo. With a median follow-up of 50 months, the 4.5-year PFS rate was 68% in the ibrutinib-and-rituximab arm (95% CI, 55%–78%) and 25% in the rituximab-and-placebo arm (95% CI, 15%–37%) (hazard ratio [HR], 0.25; 95% CI, 0.15–0.42; $P < .0001$). The OS rate at 30 months was no different in the two arms (92%–94%).[52][[Level of evidence B1](#)] Zanubrutinib was compared with ibrutinib in a randomized prospective clinical trial of 164 patients with relapsed disease and 38 with previously untreated disease.[53] With a median follow-up of 44.4 months, the PFS rate was similar in both groups at 70% to 78% (HR, 0.63; 95% CI, 0.36–1.12), and the OS rate was similar in both groups at 85% to 87% (HR, 0.75; 95% CI, 0.36–1.59).[54] The zanubrutinib group had fewer cases of atrial fibrillation (11 vs. 1) and 50% fewer cases of hypertension (statistics not provided).[53][[Level of evidence C3](#)] BTK inhibition with ibrutinib allowed all 13 patients with cold-antibody-mediated autoimmune hemolytic anemia and acrocyanosis to attain clinical remission regardless of underlying pathology or *MYD88* variant status.[55][[Level of evidence C3](#)]

Previously untreated patients who received rituximab had response rates of 60% to 80%, but close monitoring of the serum IgM is required because of a sudden rise in this paraprotein at the start of therapy.[56-58][[Level of evidence C3](#)] The rise of IgM after rituximab can be avoided with the concomitant use of an alkylating agent, such as cyclophosphamide, or the proteasome inhibitors bortezomib or ixazomib.[44,59-61] A combination of bortezomib, dexamethasone, and rituximab has been used without causing IgM rebound.[62-64] Previously untreated patients with lymphoplasmacytic lymphoma who received the nucleoside analogues cladribine and fludarabine showed similar response rates.[51,65,66][[Level of evidence C3](#)] Patients who received single-agent alkylators, bendamustine, bortezomib, venetoclax, and combination chemotherapy with or without rituximab also showed similar response rates.[51,59,61,67-71]

[[Level of evidence C3](#)] In the rare case of lymphoplasmacytic lymphoma involving the CNS (Bing-Neel syndrome), ibrutinib resulted in an 85% response rate in an anecdotal series of 28 patients.[[72](#)][[Level of evidence C3](#)]

Myeloablative therapy with autologous or allogeneic hematopoietic stem cell support is under clinical evaluation.[[73-76](#)] Candidates for this approach should avoid long-term use of alkylating agents or purine nucleoside analogues, which can deplete hematopoietic stem cells or predispose patients to myelodysplasia or acute leukemia.[[56,77](#)] After relapse from alkylating-agent therapy, 92 patients with lymphoplasmacytic lymphoma were randomly assigned to receive either fludarabine or cyclophosphamide, doxorubicin, and prednisone. Although relapse-free survival favored fludarabine (median duration of 19 months vs. 3 months; $P < .01$), no difference was observed in OS.[[78](#)][[Level of evidence B1](#)]

Marginal Zone Lymphoma

When marginal zone lymphomas involve the nodes, they are called monocytoid B-cell lymphomas or nodal marginal zone B-cell lymphomas. When they involve extranodal sites (e.g., gastrointestinal tract, thyroid, lung, breast, orbit, and skin), they are called mucosa-associated lymphatic tissue (MALT) lymphomas.[[79,80](#)] Splenic marginal zone lymphoma is a distinct clinical entity, which usually presents with massive splenomegaly. A variant form of MALT lymphoma is known as immunoproliferative small intestinal disease (IPSID).[[80](#)] A prognostic index for all of the marginal zone lymphomas has three adverse prognostic factors: aged 70 years or older, stage III or stage IV disease, and high LDH level.[[81](#)] Fewer than 10% of patients transform to a higher-grade lymphoma. In one retrospective review, risk factors for transformation included elevated LDH, more than four nodal sites at the time of initial diagnosis of marginal zone lymphoma, and failure to achieve complete response after initial treatment.[[82](#)]

Gastric MALT

Many patients have a history of autoimmune disease, such as Hashimoto thyroiditis or Sjögren syndrome, or of *Helicobacter* gastritis. Most patients present with stage I or stage II extranodal disease, which is most often in the stomach. Treatment of *H. pylori* infection may resolve most cases of localized gastric involvement.[[83,84](#)] After standard antibiotic regimens, 50% of patients show resolution of gastric MALT by endoscopy after 3 months. Other patients may show resolution after 12 to 18 months of observation. Of the patients who attain complete remission, 30% demonstrate monoclonality by immunoglobulin heavy chain rearrangement on stomach biopsies with a 5-year median follow-up.[[85](#)] The clinical implication of this finding is unknown. Translocation t(11;18) in patients with gastric MALT predicts poor response to both antibiotic therapy and oral alkylator therapy, and predicts negative *H. pylori* testing results.[[86-88](#)] Patients with stable asymptomatic disease and persistently positive biopsies have been successfully followed with a watchful waiting approach until disease progression.[[84](#)] Patients

with disease progression are treated with radiation therapy,[89-94] rituximab,[95] surgery (total gastrectomy or partial gastrectomy plus radiation therapy),[96] chemotherapy,[97] or combined-modality therapy.[98] A single-arm prospective trial enrolled 24 patients with newly diagnosed or relapsed *H. pylori*-negative gastric MALT.[94] Most patients had a complete response to radiation therapy at 4 Gy, and 20 Gy was applied as salvage therapy. The 3-year local control rate was 96% (95% CI, 88%–100%).[94][[Level of evidence C3](#)] Endoscopic ultrasonography may help clinicians monitor responses in these patients.[99] Four case series encompassing more than 100 patients with stage IE or IIE DLBCL with or without associated MALT (but *H. pylori*-positive) reported durable complete remissions in more than 50% of the patients after treatment of *H. pylori*.[\[100-103\]](#)

Extragastric MALT

Localized involvement of other sites can be treated with radiation therapy or surgery.[\[90-92,104-107\]](#) Patients with extragastric MALT lymphoma have a higher relapse rate than patients with gastric MALT lymphoma in some series, and relapses can happen many years and even decades later.[\[108\]](#) Many of these recurrences involve different MALT sites than the original location.[\[109\]](#) When disseminated to lymph nodes, bone marrow, or blood, this entity behaves like other low-grade lymphomas.[\[110,111\]](#) A prospective randomized trial of 401 patients with nongastric extranodal MALT compared chlorambucil alone versus rituximab plus chlorambucil versus rituximab alone.[\[112\]](#) With a median follow-up of 7.4 years, the event-free survival was 68% in the rituximab-plus-chlorambucil arm, 51% in the rituximab-alone arm, and 50% in the chlorambucil-alone arm ($P = .0009$). However, the 5-year OS rate was 90% in all arms.[\[112\]](#) For patients with ocular adnexal MALT, antibiotic therapy using doxycycline that targeted *Chlamydia psittaci* resulted in durable remissions for almost one-half of the patients in a review of the literature that included 131 patients.[\[113\]](#)[\[Level of evidence C3\]](#) These responses to doxycycline are mainly seen in Italian trials and less often in trials conducted in other geographic sites.[\[114\]](#) Large B-cell lymphomas of MALT sites are classified and treated as diffuse large cell lymphomas.[\[115\]](#) A large, retrospective review of primary ocular adnexal MALT found that after 10 years of follow-up, 4% of stage I patients treated with radiation therapy had disease transformation to DLBCL, and 3% developed CNS involvement.[\[116\]](#)

Nodal marginal zone lymphoma

Patients with nodal marginal zone lymphoma (monocytoid B-cell lymphoma) are treated with watchful waiting or therapies as described for lymphoplasmacytic lymphoma. Rituximab alone, obinutuzumab alone, or combinations with cytotoxic agents (such as bendamustine, CVP [cyclophosphamide, vincristine, and prednisone] or CHOP [cyclophosphamide, doxorubicin, vincristine, and prednisone]) can be used.[\[117-119\]](#)[\[Level of evidence C3\]](#) Zanubrutinib is approved for patients with disease relapse after a rituximab-containing regimen. This approval was based on a single-arm phase II study.[\[120\]](#) With a median follow-up of 15.7 months, the overall response rate was 68.2%, and the complete response rate was 25.8%. The median

duration of response was 93% at 12 months.[120][[Level of evidence C3](#)] Ibrutinib also showed similar efficacy in patients with relapsed marginal zone lymphoma.[121][[Level of evidence C3](#)] Patients with marginal zone lymphoma comprised a small percentage (about 15%) of the population for a trial that established the benefit of rituximab with lenalidomide in patients with relapsed or refractory indolent lymphoma (AUGMENT trial).[122][[Level of evidence C3](#)] Similar to follicular lymphoma, patients with POD24 who required initiation of therapy had a worse prognosis than did the patients without POD24 (3-year OS rate, 53% vs. 95%).[123] Among patients with concomitant HCV infection, 40% to 60% attained a complete or partial remission after loss of detectable HCV RNA with antiviral treatment.[124,125][[Level of evidence C3](#)]

Mediterranean abdominal lymphoma

The disease variously known as Mediterranean abdominal lymphoma, heavy-chain disease, or IPSID, which occurs in young adults in eastern Mediterranean countries, is another version of MALT lymphoma. This disease responds to antibiotics in its early stages.[126] *Campylobacter jejuni* has been identified as one of the bacterial species associated with IPSID, and antibiotic therapy may result in disease remission.[127]

Splenic marginal zone lymphoma

Splenic marginal zone lymphoma is an indolent lymphoma that is marked by massive splenomegaly and peripheral blood and bone marrow involvement, usually without adenopathy.[128,129] This type of lymphoma is also known as splenic lymphoma with villous lymphocytes. Splenectomy may result in prolonged remission.[130,131]

Management is similar to that of other low-grade lymphomas and usually involves rituximab alone or rituximab in combination with purine analogues or alkylating agent chemotherapy.[132] Splenic marginal zone lymphoma responds less well to chemotherapy, which would ordinarily be effective for chronic lymphocytic leukemia.[128,132,133] Among small numbers of patients with splenic marginal zone lymphoma (splenic lymphoma with villous lymphocytes) and HCV infection, most attained a complete or partial remission after loss of detectable HCV RNA with treatment using interferon alfa with or without ribavirin.[124,134]; [135][[Level of evidence C3](#)] In contrast, no responses to interferon were seen in six HCV-negative patients.

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Treatment Option Overview for Indolent B-Cell Non-Hodgkin Lymphoma

Treatment of indolent non-Hodgkin lymphoma (NHL) depends on the histological type and stage. Many of the improvements in survival have been made because of clinical trials that have attempted to improve on conventional or standard therapy.

In asymptomatic patients with indolent forms of advanced NHL, treatment may be deferred until the patient becomes symptomatic as the disease progresses. When treatment is deferred, the clinical course of patients with indolent NHL varies; frequent and careful observation is required so that effective treatment can be initiated when the clinical course of the disease accelerates. Some patients have a prolonged indolent course, but others have disease that rapidly evolves into more aggressive types of NHL that require immediate treatment.

Radiation techniques differ somewhat from those used in the treatment of Hodgkin lymphoma. The dose of radiation therapy usually varies from 25 Gy to 50 Gy and is dependent on factors that include the histological type of lymphoma, the patient's stage and overall condition, the goal of treatment (curative or palliative), the proximity of sensitive surrounding organs, and whether the patient is being treated with radiation therapy alone or in combination with chemotherapy. Given the patterns of disease presentations and relapse, treatment may need to include unusual sites such as Waldeyer ring, epitrochlear nodes, or mesenteric nodes. The associated morbidity of the treatment must be considered carefully. Most patients who receive radiation are treated on only one side of the diaphragm. Localized presentations of extranodal NHL may be treated with involved-field techniques with significant (>50%) success.

In situations where mediastinal radiation therapy would encompass the left side of the heart or would increase breast cancer risk in young female patients, proton therapy may be considered to reduce radiation dose to organs at risk.^[1] For more information, see the [Superior Vena Cava Syndrome](#) section in [Cardiopulmonary Syndromes](#).

Table 4. Treatment Options for Indolent B-Cell Non-Hodgkin Lymphoma (NHL)

Stage	Treatment Options
Indolent Stage I and Indolent, Contiguous Stage II B-cell NHL	Radiation therapy
	Rituximab with or without chemotherapy
	Watchful waiting
	Other therapies as designated for patients with advanced-stage disease
Indolent, Noncontiguous Stage II/III/IV B-cell NHL	Watchful waiting for asymptomatic patients
	Rituximab alone or in combination with cytotoxic agents used in front-line therapy
	Lenalidomide and rituximab
	Maintenance rituximab
	Obinutuzumab alone or in combination with cytotoxic agents used in front-line therapy
Indolent, Recurrent B-cell NHL	Rituximab alone or in combination with cytotoxic agents used in front-line therapy
	Obinutuzumab alone or in combination with cytotoxic agents used in front-line therapy
	Lenalidomide and rituximab
CAR = chimeric antigen receptor; SCT = stem cell transplant.	

Stage	Treatment Options
	Zanubrutinib and obinutuzumab
	EZH2 inhibitor
	Bispecific T-cell engagers
	CAR T-cell therapy
	SCT
	Palliative radiation therapy
CAR = chimeric antigen receptor; SCT = stem cell transplant.	

Even though existing treatments cure a significant fraction of patients with lymphoma, numerous clinical trials that explore treatment improvements are in progress. If possible, patients can be included in these studies. Standardized guidelines for response assessment have been suggested for use in clinical trials.[2]

Several retrospective reviews suggest that routine surveillance scans offer little to no value in patients with diffuse-large B-cell lymphoma (DLBCL) who have attained a clinical complete remission after induction therapy. Prognostic value is also difficult to identify for an interim positron emission tomography–computed tomography scan during induction therapy for DLBCL.[3-6]

Active hepatitis B or hepatitis C can be assessed before treatment with rituximab and/or chemotherapy.[7,8] Patients with detectable hepatitis B virus (HBV) benefit from prophylaxis with entecavir in the context of rituximab therapy.[9,10] Patients with a resolved HBV infection (defined as hepatitis B surface antigen-negative but hepatitis B core antibody-positive) are at risk of reactivation of HBV and require monitoring of HBV DNA. In a retrospective study of 326 patients, prophylactic nucleoside analogue therapy lowered HBV reactivation from 10.8% to 2.1%.[11] Similarly, prophylaxis for herpes zoster with acyclovir or valacyclovir and prophylaxis for *Pneumocystis* with trimethoprim/sulfamethoxazole or dapsone are usually given to patients receiving rituximab with or without combination chemotherapy. Long-term impaired immune health was evaluated in a retrospective cohort study of 21,690 survivors of DLBCL from the California Cancer Registry. Elevated incidence rate ratios were found up to 10 years later for

pneumonia (10.8-fold), meningitis (5.3-fold), immunoglobulin deficiency (17.6-fold), and autoimmune cytopenias (12-fold).[12]

Among 2,508 patients in a Danish registry, the incidence of doxorubicin-induced congestive heart failure increased for 115 NHL survivors with a history of cardiac disease (hazard ratio [HR], 2.71; 95% confidence interval [CI], 1.15–6.36) and/or multiple cardiovascular risk factors (HR, 2.86; 95% CI, 1.56–5.23).[13]

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Treatment of Indolent Stage I and Indolent, Contiguous Stage II B-Cell Non-Hodgkin Lymphoma

Although localized presentations are uncommon in B-cell non-Hodgkin lymphoma (NHL), the goal of treatment is to cure the disease in patients with confirmed localized occurrence after undergoing appropriate staging.

Treatment Options for Indolent Stage I and Indolent, Contiguous Stage II B-Cell NHL

Treatment options for indolent stage I and indolent, contiguous stage II B-cell NHL include:

1. [Radiation therapy.](#)
2. [Rituximab with or without chemotherapy.](#)
3. [Watchful waiting.](#)
4. [Other therapies, as designated for patients with advanced-stage disease.](#)

In a prospective randomized trial, 150 patients with stage I or stage II follicular lymphoma were randomly assigned to 30 Gy of involved-field radiation therapy alone or radiation therapy plus six cycles of R-CVP (rituximab, cyclophosphamide, vincristine, prednisolone). With a median follow-up of 9.6 years, the 10-year progression-free survival (PFS) rate favored combined-modality therapy, at 59% (95% confidence interval [CI], 46%–74%) versus 41% for radiation therapy alone (95% CI, 30%–57%) ($P = .033$). There was no difference in overall survival (OS) (87% and 95%; $P = .40$).^[1][\[Level of evidence B1\]](#)

The National Lymphocare Study identified 471 patients with stage I follicular lymphoma. Of those patients, 206 were rigorously staged with a bone marrow aspirate and biopsy, and computed tomography (CT) scans or positron emission tomography (PET)-CT scans.[2] Nonrandomized treatments included radiation therapy (27%), rituximab-chemotherapy (R-chemotherapy) (28%), watchful waiting (17%), R-chemotherapy plus radiation therapy (13%), and rituximab alone (12%), although more than one-third of the patients started with watchful waiting. With a median follow-up of 57 months, PFS favored R-chemotherapy or R-chemotherapy plus radiation therapy, but OS was nearly identical, all over 90%. [2][[Level of evidence C2](#)] Clinical trials are required to answer the following questions:[3]

- If the PET-CT scan is clear after excisional biopsy, is watchful waiting or radiation therapy preferred?
- Should rituximab be added to radiation therapy for stage I follicular lymphoma?
- Is there any role for R-chemotherapy plus radiation therapy?

Radiation therapy

Long-term disease control within radiation fields can be achieved in a significant number of patients with indolent stage I or stage II NHL by using radiation therapy. This requires dosages of radiation that usually range from 25 Gy to 40 Gy to involved sites or to extended fields that cover adjacent nodal sites.[1,4-6] Almost one-half of all patients treated with radiation therapy alone will have out-of-field relapse within 10 years.[1,6,7]

A retrospective review of 512 patients from an international consortium evaluated patients with early-stage follicular lymphoma who received at least 24 Gy of localized radiation therapy at initial presentation. With a median follow-up of 52 months, 29.1% of patients developed recurrent lymphoma at a median of 23 months (range, 1–143 months).[8] With a median follow-up of 33 months after relapse, the 3-year OS rate was 91.4% after patients received subsequent systemic chemotherapy that involved rituximab.[8]

Very low-dose radiation therapy with 4 Gy (2 Gy × 2 fractions) can result in 50% remission rates for patients who cannot tolerate higher doses.[9] In a multicenter, randomized, prospective trial, 548 patients with follicular or marginal zone lymphoma received radiation therapy, either 4 Gy in 2 fractions or 24 Gy in 12 fractions.[10]

- At a median follow-up of 73.8 months, the 5-year local complete response rate was 89.9% (85.5%–93.1%) after 24 Gy and 70.4% (64.7%–75.4%) after 4 Gy (hazard ratio, 3.46; 95% CI, 2.25–5.33; $P < .0001$). [10]
- Although durable local control was superior for patients who received 24 Gy, the 4 Gy regimen was nearly comparable with reductions in radiation exposure, time undergoing therapy, and cost.

Rituximab with or without chemotherapy

When radiation therapy is contraindicated, or when an alternative treatment is preferred, patients with symptomatic disease who require therapy may receive rituximab with or without chemotherapy (as outlined [below](#) for more advanced-stage patients).[11] The value of adjuvant treatment with radiation to decrease relapse, plus rituximab (an anti-CD20 monoclonal antibody) either alone or in combination with chemotherapy, has been extrapolated from trials of patients with advanced-stage disease and has not been confirmed.[12,13]

Watchful waiting

Watchful waiting can be considered for asymptomatic patients.[14] Watchful waiting has never been compared with up-front radiation therapy in a prospective randomized trial. A retrospective analysis of the Surveillance, Epidemiology and End Results (SEER) Program database in patients diagnosed over a span of 30 years showed improved outcomes for up-front radiation therapy.[15]

Other therapies as designated for patients with advanced-stage disease

Patients with disease unable to be encompassed by radiation therapy are treated as outlined for patients with stage III or stage IV low-grade lymphoma.

Current Clinical Trials

Use our [advanced clinical trial search](#) to find NCI-supported cancer clinical trials that are now enrolling patients. The search can be narrowed by location of the trial, type of treatment, name of the drug, and other criteria. [General information](#) about clinical trials is also available.

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Treatment of Indolent, Noncontiguous Stage II/III/IV B-Cell Non-Hodgkin Lymphoma

Optimal treatment of advanced stages of low-grade non-Hodgkin lymphoma (NHL) is controversial because current therapeutic options result in low cure rates. Numerous clinical trials are in progress to evaluate treatment issues, and patients are urged to participate. The rate of relapse is fairly constant over time, even in patients who have achieved complete response to treatment. Relapse may occur many years after treatment. Currently, no randomized trials provide guidance to clinicians about the initial choice of watchful waiting, rituximab, nucleoside analogues, alkylating agents, combination chemotherapy, radiolabeled monoclonal antibodies, or combinations of these options.[1]; [2][[Level of evidence B1](#)]

For patients with indolent, noncontiguous stage II and stage III NHL, central lymphatic radiation therapy has been proposed but is not usually recommended as a form of treatment.[3,4]

Patients with a resolved hepatitis B virus (HBV) infection (defined as hepatitis B surface antigen-negative but hepatitis B core antibody-positive) are at risk of reactivation of HBV and require monitoring of HBV DNA. In a retrospective study of 326 patients, prophylactic nucleoside analogue therapy lowered HBV reactivation from 10.8% to 2.1%.[5]

Treatment Options for Indolent, Noncontiguous Stage II/III/IV B-Cell NHL

Treatment options for indolent, noncontiguous stage II/III/IV B-cell NHL include:

1. [Watchful waiting for asymptomatic patients.](#)
2. [Rituximab alone or in combination with cytotoxic agents used in front-line therapy.](#)
3. [Lenalidomide and rituximab.](#)
4. [Maintenance rituximab.](#)
5. [Obinutuzumab alone or in combination with cytotoxic agents used in front-line therapy.](#)

Because none of the therapies listed above are curative for advanced-stage disease, innovative approaches are under clinical evaluation.

Watchful waiting for asymptomatic patients

Because relapse may occur many years after treatment, even in patients who have achieved complete responses, deferred treatment (i.e., watchful waiting until the patient becomes symptomatic before initiating treatment) can be considered.[2,6-8] The Follicular Lymphoma International Prognostic Index (FLIPI) and the revised FLIPI-2 can predict progression-free survival (PFS) and overall survival (OS), but the scores cannot be used to establish the need for therapy in asymptomatic patients.[9,10]

Evidence (watchful waiting):

1. Three randomized trials compared watchful waiting with immediate chemotherapy.[7,11]; [12][Level of evidence A1]
 - All three trials showed no difference in cause-specific survival or OS.
 - For patients randomly assigned to watchful waiting, the median time to require therapy was 2 to 3 years. One-third of patients undergoing watchful waiting never required treatment (one-half died of other causes and the other half remained progression free after 10 years).
2. A selected group of 107 patients with advanced-stage follicular lymphoma were managed with initial watchful waiting; subsequent therapy was delayed for a median of 55 months. These patients achieved equivalent freedom from treatment failure and OS compared with a similar cohort treated immediately with rituximab.[13][Level of evidence C2] This implies that watchful waiting remains a relevant approach even in the rituximab era.

Rituximab alone or in combination with cytotoxic agents used in front-line therapy

Standard therapy includes rituximab, an anti-CD20 monoclonal antibody, either alone, as was shown in the [ECOG-E4402](#) trial (NCT00075946),[14-19] or in combination with purine nucleoside analogues, such as fludarabine or cladribine, alkylating agents (with or without steroids), or combination chemotherapy. Rituximab may be considered as first-line therapy, either alone or in combination with other agents. Rituximab may be given intravenously (IV) or subcutaneously (SQ), and biosimilar versions, such as CT-P10 and GP2013, have shown equivalent efficacy and safety.[20-22] Combinations include the following:

- R-bendamustine: rituximab + bendamustine.[23-25]
- R-F: rituximab + fludarabine.[26]
- R-CVP: rituximab + cyclophosphamide + vincristine + prednisone.[27-30]
- R-CHOP: rituximab + cyclophosphamide + doxorubicin + vincristine + prednisone.[29-33] A Cochrane meta-analysis could not identify any OS benefit of adding doxorubicin to chemotherapy regimens with rituximab or to chemotherapy regimens without rituximab. [34][Level of evidence A1]
- R-FM: rituximab + fludarabine + mitoxantrone.[29,30,35]
- R-FCM: rituximab + fludarabine + cyclophosphamide + mitoxantrone.[36]

Evidence (rituximab with or without chemotherapy):

1. A prospective randomized trial of 534 patients with previously untreated, advanced-stage, follicular lymphoma compared R-CHOP, R-FM, and R-CVP.[29]

- With a median follow-up of 84 months, there was no difference in OS (8-year OS rate, 83%; 95% confidence interval [CI], 79%–87%), but the 8-year PFS rates favored R-CHOP (52%) and R-FM (49%) over R-CVP (42%) (P for the three regimens = .037). [29][[Level of evidence B1](#)]

2. Four randomized prospective studies of previously untreated patients (involving more than 1,300 patients) and one Cochrane meta-analysis that included both untreated and previously treated patients (involving almost 1,000 patients) have compared rituximab plus combination chemotherapy with chemotherapy alone.[28,33,37]; [38][[Level of evidence A1](#)]

- Rituximab plus chemotherapy was superior in terms of event-free survival (EFS) or PFS (ranging from 2–3 years) in all of the studies and in terms of OS in all but one study (absolute benefit ranging from 6%–13% at 4 years, $P < .04$; hazard ratio [HR], 0.63 [0.51–0.79] for the meta-analysis).
- All of these trials were performed in symptomatic patients who required therapy. These results do not negate watchful waiting when appropriate.
- Fluorine F 18-fludeoxyglucose positron emission tomography-computed tomography (18F-FDG PET-CT) scan status at the completion of rituximab plus chemotherapy induction therapy is strongly predictive of outcome. It is not yet known whether acting on the results of the scans translates into better outcomes. [39,40]

3. In a prospective randomized trial ([NCT00991211](#)), 527 patients with indolent and mantle cell lymphoma were randomly assigned to receive either bendamustine and rituximab or R-CHOP.[24][[Level of evidence B1](#)]

- With a median follow-up of 45 months, the median PFS favored the bendamustine arm (69 months vs. 31 months [HR, 0.58; 95% CI, 0.44–0.74; $P < .0001$]) but with no difference in OS.
- Compared with the R-CHOP arm, the bendamustine arm was associated with significantly lower rates of alopecia, hematologic toxicity, stomatitis, peripheral neuropathy, and infections.

4. In a similar prospective randomized trial, 447 patients with indolent and mantle cell lymphoma were assigned to bendamustine and rituximab versus R-CHOP or R-CVP.[25] [[Level of evidence B1](#)]

- With a median follow-up of 65 months, the 5-year PFS rate favored bendamustine and rituximab, 65.5% versus 55.8% (HR, 0.61; 95% CI, 0.45–0.85; $P = .0025$), but with no difference in OS.

- Increased deaths in the bendamustine-and-rituximab arm from cardiovascular causes (seven vs. one) and from secondary malignancies other than lymphoma (five vs. three) may have contributed to the lack of OS advantage.

Lenalidomide and rituximab

The combination of the immunomodulating agent lenalidomide with rituximab (the so-called R² regimen) has been proposed as an alternative regimen to combinations involving cytotoxic agents and their subsequent short- and long-term toxicities.

Evidence (lenalidomide and rituximab):

1. In a randomized prospective trial ([RELEVANCE](#) [NCT01650701]) of 1,030 patients with previously untreated follicular lymphoma, rituximab plus lenalidomide for 18 months was compared with rituximab plus chemotherapy (usually R-CHOP).[\[41,42\]](#) All patients received rituximab maintenance for up to 2 years.
 - With a median follow-up of 72 months, the 6-year PFS rates (60% and 59%) and 3-year OS rates (89%) were identical (HR for PFS, 1.03; 95% CI, 0.84–1.27; $P = .78$) (HR for OS was not reported).[\[41,42\]](#)[\[Level of evidence A1\]](#)
 - This trial established that the R² regimen is as effective as rituximab plus cytotoxic chemotherapy options. The transformation rate to aggressive lymphoma per year was 0.68% in the R² group and 0.45% in the R-chemotherapy group. With a median follow-up of 72 months, there were no new safety signals.[\[42\]](#)
2. In a randomized prospective trial of 358 patients with resistant/refractory indolent lymphoma (usually follicular lymphoma), the R² regimen was compared with rituximab alone.[\[43\]](#)
 - With a median follow-up of 28 months, the median PFS was 39.4 months for R² and 14.1 months for rituximab alone ($P < .0001$), with no difference in OS.[\[43\]](#)[\[Level of evidence B1\]](#)

Maintenance rituximab

After induction therapy with rituximab only or with rituximab plus chemotherapy, rituximab can be used once every 2 to 3 months as maintenance therapy. Several studies have evaluated this approach.

Evidence (maintenance rituximab **for previously untreated patients**):

1. In the [PRIMA](#) study (NCT00140582), 1,018 patients with high-risk, previously untreated, symptomatic disease achieved complete response or partial response after induction therapy with immunochemotherapy (usually R-CHOP). Patients were then randomly

assigned to 2 years of maintenance rituximab versus no maintenance.[44][[Level of evidence B1](#)]

- With a median follow-up of 9.0 years, median PFS favored rituximab maintenance (10.5 years) compared with observation (4.1 years) (HR, 0.61; 95% CI, 0.52–0.73; $P < .001$), but with no difference in OS.

2. In the [United Kingdom/International Study](#) (NCT00112931), 379 previously untreated patients with asymptomatic, low-burden disease were randomly assigned to watchful waiting versus rituximab induction only versus rituximab induction followed by 2 years of rituximab maintenance.[45][[Level of evidence A3](#)]

- Although OS and histological transformation rates were not different at 3 years, maintenance rituximab was favored based on quality-of-life studies (Mental Adjustment to Cancer Scale $P = .0004$ at 7 months; Illness Coping Score $P = .0012$ at 7 months) and time-to-initiation of new treatment by 3 years (54% for watchful waiting vs. 12% for rituximab maintenance [HR, 0.21; 95% CI, 0.14–0.31; $P < .0001$]). [45][[Level of evidence A3](#)]
- This study suggested that for some patients, watch and wait resulted in watch and worry.[46] However, from the perspective of OS and histological transformation rates, no benefit could be seen with rituximab maintenance.

3. In the [RESORT](#) study (NCT00075946), 289 previously untreated patients with asymptomatic, low-burden disease were randomly assigned to receive rituximab induction alone, with a re-treatment strategy that used rituximab at relapse versus rituximab induction plus maintenance rituximab every 13 weeks until treatment failure. [47][[Level of evidence B3](#)]

- With a median follow-up of 8.7 years, the 7-year rate for freedom from cytotoxic chemotherapy or radiation therapy was 83% for patients who received maintenance rituximab and 63% for patients who received induction alone (HR, 2.37; 95% CI, 1.5–3.8; $P = .0001$). There was no difference in the 10-year OS rate (83% vs. 84%; nonsignificant $P = .5972$). A re-treatment strategy achieved comparable disease control using significantly fewer doses of rituximab.
- Maintenance rituximab induces prolonged B-cell depletion, but morbidity from infections was not evaluated after 2015, when the full protocol data stopped (12 years after the study began).

These three randomized trials in previously untreated patients showed no advantage for the use of rituximab maintenance versus observation and reinduction of therapy at the time of relapse. The trials suggest a benefit for maintenance rituximab after reinduction for relapsed disease. Many questions remain about rituximab maintenance, particularly about truncating therapy at 2 years and long-term safety and efficacy. A trial extending

rituximab maintenance to 5 years showed similar EFS or OS versus 1 year of maintenance after induction therapy with rituximab in previously untreated patients.[48][[Level of evidence A1](#)]

4. The [FOLL12](#) study (NCT02063685) included 807 patients with previously untreated high-tumor burden follicular lymphoma. Patients received rituximab plus chemotherapy induction and were randomly assigned to either standard rituximab maintenance (every 8 weeks for 2 years) or to postinduction treatment (monitoring, rituximab maintenance, or radioimmunotherapy) based on their complete metabolic response and measurable residual disease (MRD)–negative status.[49]
 - With a median follow-up of 53 months, the 3-year PFS rate was 86% for patients who received standard maintenance and 72% for patients who received response-based treatment ($P < .001$). The 3-year OS rate was the same in both groups (98% vs. 97%; $P = .238$).[49][[Level of evidence B1](#)]
 - This trial does not support the use of an end-of-treatment PET-CT scan to guide the use of maintenance rituximab.
5. A prospective trial included 202 patients with previously untreated low-tumor burden follicular lymphoma. Patients were randomly assigned to receive either four weekly doses of IV rituximab (standard dose, 375 mg/m^2) or one dose of IV rituximab followed by three weekly doses of SQ rituximab (1,400 mg) and maintenance doses in months 3, 5, 7, and 9. [19]
 - With a median follow-up of 50.2 months, the 4-year PFS rate was 58.1% (95% CI, 47.5%–67.4%) for patients in the SQ arm who received maintenance therapy, and 41.2% (95% CI, 30.6%–51.6%) (HR, 0.585; 0.939–0.871; $P = .0076$) for the patients in the IV arm who did not receive maintenance therapy.[19][[Level of evidence B1](#)]
 - There was no difference in OS or time-to-next treatment.
6. Outside the context of clinical trials, the use of MRD testing has not been shown to be predictive in directing therapy for patients with follicular lymphoma. In retrospective analyses of two randomized prospective trials, while MRD negativity was prognostic of outcome, maintenance rituximab or obinutuzumab prolonged PFS the most among patients with MRD-negative disease.[49,50][[Level of evidence C2](#)] Stopping maintenance rituximab or obinutuzumab was not indicated in patients with MRD-negative disease, negating any possible change in therapy based on that status.

In summary, for previously untreated patients, all of the studies showed improved PFS, with no change in OS.

Evidence (maintenance rituximab **for previously treated patients**):

1. In a prospective randomized trial of 465 patients with relapsed follicular lymphoma, responders to R-CHOP or CHOP were further randomly assigned to receive rituximab maintenance (1 dose every 3 months for 2 years) or no maintenance.[51][[Level of evidence B1](#)]
 - At a median follow-up of 6 years, rituximab maintenance was better for median PFS (44 months vs. 16 months, $P < .001$) and borderline for 5-year OS (74% vs. 64%, $P = .07$).
 - This benefit for maintenance was evident even for patients who received rituximab during induction therapy. Most patients in both arms received extensive rituximab during postprotocol salvage treatment.
2. In a prospective randomized trial of 280 patients with relapsed follicular lymphoma, responders to chemotherapy and autologous stem cell transplant consolidation were randomly assigned to receive four doses of rituximab maintenance or no maintenance. [52][[Level of evidence B1](#)]
 - With an 8.3-year median follow-up, the 10-year PFS rates favored maintenance (54% vs. 37% [HR, 0.66; 95% CI, 0.47–0.91; $P = .012$]), but there was no difference in OS.
3. A meta-analysis of nine randomized clinical trials with a total of 2,586 patients with follicular lymphoma, most of whom had relapsed disease, compared rituximab maintenance with no maintenance and showed improved OS for rituximab maintenance in previously treated patients (HR_{death}, 0.72; 95% CI, 0.57–0.91).[53][[Level of evidence A1](#)]

For previously treated patients, there is more evidence to suggest an OS advantage with the use of rituximab maintenance.

Obinutuzumab alone or in combination with cytotoxic agents used in front-line therapy

Obinutuzumab is a glycoengineered type II anti-CD20 monoclonal antibody with greater antibody-dependent cellular cytotoxicity than rituximab.

Evidence (obinutuzumab):

1. A prospective randomized trial ([NCT01332968](#)) of 1,202 patients with previously untreated follicular lymphoma compared obinutuzumab combined with bendamustine (50%), CHOP (cyclophosphamide, doxorubicin, vincristine, and prednisone) (33%), or CVP (cyclophosphamide, vincristine, and prednisone) (10%) with rituximab combined with the same chemotherapy regimens (based on investigator choice).[54] After six cycles of

combination therapy, patients had 2 years of maintenance therapy, receiving the same antibody every 2 months.

- With a median follow-up of 34.5 months, the 3-year PFS rate was 80% in the obinutuzumab group and 73.3% in the rituximab group (HR, 0.66; 95% CI, 0.51–0.85; $P = .001$).[\[54\]](#)[\[Level of evidence B1\]](#)
- There was no difference in OS.
- Compared with what has been seen historically, there was a high rate of toxic deaths among patients using bendamustine in the obinutuzumab arm (5.6%) and in the rituximab arm (4.4%). For patients with indolent low-grade lymphoma, with median survivals exceeding 15 years, the number of toxic deaths during first-line therapy seems excessive. By comparison, the toxic death rate was 1% to 2% when either antibody was combined with CHOP or CVP.

Several issues have been raised about this study:

- The side effects were significantly higher with obinutuzumab in terms of infusion reactions and subsequent adverse events.
- Obinutuzumab costs significantly more than rituximab.

In summary, in the absence of any change in OS, switching from rituximab to obinutuzumab in combination with chemotherapy for previously untreated follicular lymphoma is a difficult choice. The PFS differences may be attributable to the imbalance in monoclonal antibody dosing, and the increased side effects and costs are mitigating factors. In this trial, bendamustine combined with either antibody led to unacceptable rates of toxic death.

Current Clinical Trials

Use our [advanced clinical trial search](#) to find NCI-supported cancer clinical trials that are now enrolling patients. The search can be narrowed by location of the trial, type of treatment, name of the drug, and other criteria. [General information](#) about clinical trials is also available.

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Treatment of Indolent, Recurrent B-Cell Non-Hodgkin Lymphoma

In general, treatment with standard agents rarely produces a cure in patients with relapsed B-cell non-Hodgkin Lymphoma (NHL). Sustained remissions after relapse can often be obtained in patients with indolent lymphomas, but relapse will usually ensue. Favorable survival after relapse has been associated with an age younger than 60 years, complete remission rather than partial remission, and duration of response longer than 2 years.[1] Even patients in the most favorable subset, however, have a tenfold greater mortality compared with age-adjusted U.S. population rates.[2]

Patients who experience a relapse of indolent lymphoma can often achieve disease control with single-agent or combination chemotherapy, rituximab (an anti-CD20 monoclonal antibody), lenalidomide, radiolabeled anti-CD20 monoclonal antibodies, or palliative radiation therapy.[3,4] However, long-term freedom from second relapse is uncommon and multiple relapses usually occur. Patients with indolent lymphoma may experience a relapse with a more aggressive histology. If the clinical pattern of relapse suggests that the disease is behaving in a more aggressive manner, a biopsy can be performed. If a more aggressive histology is confirmed, therapy must change to a regimen applicable to that histological type.[5] Rapid growth or discordant growth between various disease sites may indicate a histological conversion.

In a retrospective review of 325 patients diagnosed between 1972 and 1999, the 10-year risk of histological transformation was 30%.[6] In this series, high-risk factors for subsequent histological transformation were advanced stage, high-risk Follicular Lymphoma International Prognostic Index, and watchful waiting (as opposed to treatment being initiated at diagnosis). The median survival after transformation was 1 to 2 years, with 25% of patients alive at 5 years and with approximately 10% to 20% of patients alive 10 years after re-treatment.[7]

A prospective trial of 631 patients with follicular lymphoma and with a median follow-up of 60 months in the rituximab era (2002–2009) found a 5-year transformation rate to a higher-grade histology of 11%.[8] The median overall survival (OS) after transformation was 50 months, and the 5-year OS rate was 66%, if the transformation occurred more than 18 months after a diagnosis of follicular lymphoma. This series described a better prognosis for patients with transformation than in the era before rituximab.

For descriptions of the regimens used to treat histological conversions, see the [Treatment of Aggressive, Recurrent B-Cell Non-Hodgkin Lymphoma](#) section in Aggressive B-Cell Non-Hodgkin Lymphoma Treatment. The durability of the second remission may be short, and clinical trials can be considered.

Treatment Options for Indolent, Recurrent B-Cell NHL

Treatment options for indolent, recurrent B-cell NHL include:

1. Rituximab alone or in combination with cytotoxic agents used in front-line therapy.
2. Obinutuzumab alone or in combination with cytotoxic agents used in front-line therapy.
3. Lenalidomide and rituximab.
4. Zanubrutinib and obinutuzumab.
5. EZH2 inhibitor.
 - Tazemetostat.
6. Bispecific T-cell engagers.
 - Mosunetuzumab.
 - Epcoritamab.
7. Chimeric antigen receptor (CAR) T-cell therapy.
8. Stem cell transplant (SCT).
9. Palliative radiation therapy.

Rituximab alone or in combination with cytotoxic agents used in front-line therapy

Rituximab results in a 40% to 50% response rate in patients with relapsed indolent B-cell lymphomas.[9-13] Rituximab can also be combined with combination chemotherapy.[14,15]

Evidence (rituximab):

1. In three randomized, prospective studies involving previously treated patients with relapsed indolent lymphoma, patients were randomly assigned to rituximab maintenance after re-treatment with combination chemotherapy (with or without rituximab during induction) or rituximab alone.[16-18]
 - All trials showed prolongation of response duration.[16-18] One trial with a median follow-up of 39 months demonstrated improvement in median progression-free survival (PFS) (3.7 years vs. 1.3 years, $P < .001$) and 5-year OS rate (74% vs. 64%, $P = .07$), favoring maintenance rituximab.[17]

Obinutuzumab alone or in combination with cytotoxic agents used in front-line therapy

Obinutuzumab is a CD20-binding monoclonal antibody with alternative epitope binding.

Evidence (obinutuzumab):

1. A randomized prospective trial ([NCT01059630](#)) included 396 patients with rituximab-refractory indolent lymphoma (mostly follicular lymphoma). Patients received obinutuzumab plus bendamustine, followed by obinutuzumab maintenance therapy, for 2 years versus bendamustine alone with no maintenance therapy.[19,20][[Level of evidence A1](#)]
 - With a median follow-up of 31.8 months, the 2-year OS rate favored the obinutuzumab combination (74.5% vs. 65.1%) (hazard ratio [HR], 0.67; 95% confidence interval [CI], 0.47–0.96; $P = .027$). The median PFS also favored the obinutuzumab combination (25.8 months [95% CI, 19.5–41.1] vs. 14.1 months [95% CI, 12.6–16.0]) (HR, 0.57; 95% CI, 0.44–0.73; $P < .001$).[20][[Level of evidence A1](#)]
 - The contribution of maintenance therapy to the outcome could not be assessed in this design.

Lenalidomide and rituximab

Responses of 20% to 56% have been reported for lenalidomide, especially in patients with follicular lymphoma and small lymphocytic lymphoma, with even higher responses noted for the combination of lenalidomide and rituximab.[21,22][[Level of evidence C3](#)]

Zanubrutinib and obinutuzumab

Evidence (zanubrutinib and obinutuzumab):

1. A randomized, multicenter, phase II study ([ROSEWOOD](#) [NCT03332017]) included 217 patients with relapsed or refractory follicular lymphoma after receiving two or more prior lines of therapy. Patients received either the oral Bruton tyrosine kinase inhibitor zanubrutinib plus obinutuzumab or obinutuzumab alone until disease progression.[23][[Level of evidence B1](#)]
 - With a median follow-up of 19.0 months, the median PFS was 28.0 months for patients who received zanubrutinib plus obinutuzumab and 10.4 months for patients who received obinutuzumab alone (HR, 0.50; 95% CI, 0.33–0.75; $P < .001$).
 - The overall response rate was 69% (95% CI, 61%–76%) in the zanubrutinib-plus-obinutuzumab arm and 46% (95% CI, 34%–58%) in the obinutuzumab-alone arm ($P = .0012$).

The U.S. Food and Drug Administration approved zanubrutinib and obinutuzumab for patients with relapsed or refractory follicular lymphoma after two or more prior lines of therapy.

EZH2 inhibitor

Tazemetostat

Tazemetostat is an inhibitor of EZH2, a histone methyltransferase essential to the formation of lymph node germinal centers, especially with activating variants of *EZH2*.

Evidence (tazemetostat):

1. A phase II study included 99 patients with relapsed or refractory follicular lymphoma, 45 of whom had activating *EZH2* variants, and 54 of whom had wild-type *EZH2*.[\[24\]](#)
 - Treatment with tazemetostat resulted in an objective response rate of 69% (95% CI, 53%–82%) for patients with activating *EZH2* variants versus 35% (95% CI, 23%–49%) for patients with wild-type *EZH2*.[\[24\]](#)
 - With a median follow-up of 22 months, the median PFS was 13.8 months (95% CI, 10.7–22.0) for patients with activating *EZH2* variants and 11.1 months (95% CI, 3.7–14.6) for patients with wild-type *EZH2*.[\[24\]](#)[\[Level of evidence C3\]](#)
 - Grade 3 or 4 treatment-related adverse events occurred in 4% of patients.

Bispecific T-cell engagers

Bispecific T-cell engagers bind to CD20 (or CD19) and to CD3 to direct T cells to eliminate malignant B cells.[\[25,26\]](#) Similar to CAR T-cell therapy, almost one-half of patients who receive this therapy experience cytokine release syndrome.

Mosunetuzumab

Mosunetuzumab is a bispecific T-cell engager that binds to CD20 and CD3.[\[25\]](#)

Evidence (mosunetuzumab):

1. A single-arm, multicenter, phase II study included 90 patients with relapsed or refractory follicular lymphoma who had received two or more prior lines of therapy (including an anti-CD20 therapy and an alkylating agent). Patients received mosunetuzumab.[\[27\]](#)
 - With a median follow-up of 37.4 months, the objective response rate was 77.8% (95% CI, 67.8%–85.9%), and the complete response rate was 60.0% (95% CI, 49.1%–70.2%), per investigator assessment. The median PFS (per investigator assessment) was 24.0 months. The 3-year OS rate was 82.4% (95% CI, 73.8%–91.0%).[\[27\]](#)[\[Level of evidence C1\]](#)
 - Cytokine release syndrome occurred in 44.4% of patients; 97.2% of cases were grade 1 or 2 in severity.

Epcoritamab

Epcoritamab is a bispecific T-cell engager that binds to CD20 and CD3.

Evidence (epcoritamab):

1. A single-arm multicenter study ([NCT03625037](#)), included 128 patients with relapsed or refractory follicular lymphoma who had received two or more lines of therapy. Patients received epcoritamab.[\[28\]](#)
 - With a median follow-up of 17.4 months, the overall response rate was 82.0 % (95% CI, 74.3%–88.3%) and the complete response rate was 62.5% (95% CI, 53.5%–70.9%).[\[28\]](#)[\[Level of evidence C3\]](#)
 - Only 2% of patients had grade 3 or higher cytokine release syndrome events.

Chimeric antigen receptor (CAR) T-cell therapy

CAR T-cell therapy, with the autologous anti-CD19 therapeutics axicabtagene ciloleucel (axi-cel), lisocabtagene maraleucel (liso-cel), or tisagenlecleucel (tisa-cel), has been approved for patients with relapsed follicular lymphoma after two or more lines of prior therapy.

Evidence (CAR T-cell therapy):

1. In a phase II trial, 159 patients with relapsed or refractory follicular lymphoma (n = 127), marginal zone lymphoma (n = 31), or diffuse large B-cell lymphoma (n = 1) received axi-cel.[\[29,30\]](#)
 - With a median follow-up of 41.7 months, the overall response rate was 92% (95% CI, 85%–97%), and the complete response rate was 74%.
 - The 18-month PFS rate was 64.8% (95% CI, 54.2%–73.5%).[\[29\]](#)[\[Level of evidence C2\]](#)
 - The median PFS was 40.2 months for patients with follicular lymphoma and was not reached for patients with marginal zone lymphoma.[\[30\]](#)[\[Level of evidence C2\]](#)
 - Cytokine release syndrome occurred in 78% of patients and was grade 3 or 4 in 6% of patients.
 - Tocilizumab was required in 50% of all patients, and 5% required vasopressors. Grade 3 or 4 neurological events occurred in 15% of patients.
2. In a phase II trial, 98 patients with relapsed or refractory follicular lymphoma after two or more lines of prior therapy received anti-CD19 CAR T-cell therapy with tisa-cel.[\[31\]](#)
 - With a median follow-up of 16.6 months, the complete response rate was 69.1% (95% CI, 58.8%–78.3%), and the overall response rate was 86.2% (95% CI, 77.5%–92.4%).[\[31\]](#)[\[Level of evidence C3\]](#)
 - Grade 3 or 4 cytokine release syndrome occurred in 48.5% of patients, and 37.1% had grade 3 or 4 neurotoxicity.
3. A phase II trial ([NCT04245839](#)) included 130 patients with relapsed or refractory follicular lymphoma who had received two or more prior lines of therapy. Patients had high-risk

features (progression of disease within 24 months after first-line chemoimmunotherapy or disease refractory to rituximab and alkylators). Patients received liso-cel.[32]

- With a median follow-up of 18.9 months, the overall response rate was 97% (95% CI, 91.6%–99.4%), and the complete response rate was 94% (95% CI, 87.5%–97.8%). [32][[Level of evidence C3](#)]
- Cytokine release syndrome occurred in 56% of patients and was grade 3 or higher in 1% of patients. Neurological side effects occurred in 15% of patients and was grade 3 or higher in 2% of patients.

CAR T cells are being used for high-risk patients whose disease has relapsed rapidly after chemoimmunotherapy. Such an approach is considered in the context of numerous other available agents.

Stem cell transplant

In many institutions, autologous or allogeneic SCTs are being used for high-risk patients whose disease has relapsed rapidly after chemoimmunotherapy. Such an approach is considered in the context of numerous other available agents.[33-37]

Evidence (SCT):

1. The German Low-Grade Lymphoma Study Group treated 307 patients with follicular lymphoma with two cycles of CHOP (cyclophosphamide, doxorubicin, vincristine, and prednisone)-like induction chemotherapy and then randomly assigned them to receive autologous SCT or interferon maintenance.[38]
 - With a median follow-up of 4.2 years, the 5-year PFS rate was 65% for patients who received a transplant versus 33% for patients who received interferon ($P < .001$). There was no difference in OS.[38][[Level of evidence B1](#)]

Palliative radiation therapy

Palliation may be achieved with very low-dose (4 Gy) involved-field radiation therapy in two fractions for patients with indolent and aggressive relapsed disease.[39] In a prospective randomized trial, treatment with 4 Gy was inferior to treatment with 24 Gy in 12 fractions in PFS (77% vs. 92%, $P < .0001$).[40][[Level of evidence B1](#)]

Current Clinical Trials

Use our [advanced clinical trial search](#) to find NCI-supported cancer clinical trials that are now enrolling patients. The search can be narrowed by location of the trial, type of treatment, name of the drug, and other criteria. [General information](#) about clinical trials is also available.

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Other Lymphoproliferative and Related Disorders

Castleman Disease

A biopsy of localized or multifocal collections of lymph nodes may lead to a diagnosis of Castleman disease (CD). Strictly speaking, this uncommon diagnosis is not a lymphoma or even a malignancy. Yet, many patients with CD may be seen and treated by hematologists or oncologists.

Localized or unicentric CD is usually asymptomatic and occurs in the mediastinum, which is the most common presentation for CD.^[1] Watchful waiting, surgery, or radiation therapy can be

used to treat this indolent form. Multicentric CD (30% of CD patients) presents with lymphadenopathy in multiple sites; symptoms such as fever, night sweats, weight loss, and fatigue; and laboratory abnormalities such as anemia, low albumin level, elevated C-reactive protein level, and high fibrinogen level.[1] Multicentric CD (MCD) is subdivided into human herpes virus-8-associated MCD (usually with HIV or severe immunocompromise) or idiopathic MCD. Cytopenias and cytokine storm are attributed to interleukin-6 (IL-6) overproduction. MCD is a feature seen in POEMS (polyneuropathy, organomegaly, endocrinopathy, monoclonal gammopathy, and skin abnormalities) syndrome [2] and TAFRO (thrombocytopenia, anasarca, fever, reticulosis, and organomegaly) syndrome.[3,4] Therapy with siltuximab (an anti-IL-6 monoclonal antibody), rituximab (an anti-CD20 monoclonal antibody), or chemotherapeutic agents has been presented in anecdotal nonrandomized series.[5-8]

True Histiocytic Lymphoma

True histiocytic lymphomas are very rare tumors that show histiocytic differentiation and express histiocytic markers in the absence of B-cell or T-cell lineage-specific immunologic markers.[9,10] Care must be taken with immunophenotypic tests to exclude anaplastic large cell lymphoma or hemophagocytic syndromes caused by viral infections, especially Epstein-Barr virus.

Therapeutic options

Therapy is modeled after the treatment of comparably staged diffuse large cell lymphomas, but the optimal approach remains to be defined.

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Latest Updates to This Summary (05/14/2025)

The PDQ cancer information summaries are reviewed regularly and updated as new information becomes available. This section describes the latest changes made to this summary as of the date above.

General Information About B-Cell Non-Hodgkin Lymphoma

Revised [text](#) to state that aggressive non-Hodgkin lymphoma (NHL) has a worse prognosis in the short term, but a significant number of patients can be cured with intensive combination chemotherapy regimens. More than 70% of patients with aggressive NHL can be cured.

Indolent B-Cell Non-Hodgkin Lymphoma

Revised [text](#) about the Follicular Lymphoma International Prognostic Index to state that patients with zero or one risk factor have a 10-year survival rate of 67%, and four or five risk factors confer a 10-year survival rate of 36%. Also added text about the nine prognostic variables identified by the Follicular Lymphoma Evaluation Index model.

Treatment of Indolent, Recurrent B-Cell Non-Hodgkin Lymphoma

Revised [text](#) about the results of a single-arm, multicenter, phase II study of 90 patients with relapsed or refractory follicular lymphoma who received mosunetuzumab (cited Sehn et al. as reference 27 and level of evidence C1).

Revised [text](#) about the results of a phase II trial of 159 patients with relapsed or refractory follicular lymphoma, marginal zone lymphoma, or diffuse large B-cell lymphoma who received axicabtagene ciloleucel (cited Neelapu et al. as reference 30 and level of evidence C2).

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About This PDQ Summary

Purpose of This Summary

This PDQ cancer information summary for health professionals provides comprehensive, peer-reviewed, evidence-based information about the treatment of indolent adult B-cell non-Hodgkin lymphoma. It is intended as a resource to inform and assist clinicians in the care of their patients. It does not provide formal guidelines or recommendations for making health care decisions.

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The lead reviewers for Indolent B-Cell Non-Hodgkin Lymphoma Treatment are:

- Eric J. Seifter, MD (Johns Hopkins University)
- Cole H. Sterling, MD (Johns Hopkins University)

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